# MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

## OVERALL SEVERITY: CRITICAL

| | |
|---|---|
| **Case ID** | `case01_baseline_vs_case03_t1059_attack` |
| **Generated** | `2026-04-19 19:11:19 UTC` |
| **Baseline capture** | `case01_baseline` |
| **Attack capture** | `case03_t1059_attack` |
| **Baseline path** | `/MFF/cases/case01_baseline` |
| **Attack path** | `/MFF/cases/case03_t1059_attack` |

# 1. EXECUTIVE SUMMARY

| Metric | Value | Metric | Value |
|---|---|---|---|
| New Processes (Attack Only) | 16 | Critical Risk Processes | 3 |
| Gone Processes (Baseline Only) | 16 | ATT&CK Techniques Observed | 2 |
| New Network Connections | 133 | ATT&CK Tactics Observed | 1 |
| Flagged Connections | 4 | IOCs Extracted | 52 |

### ATT&CK; Tactics Observed:

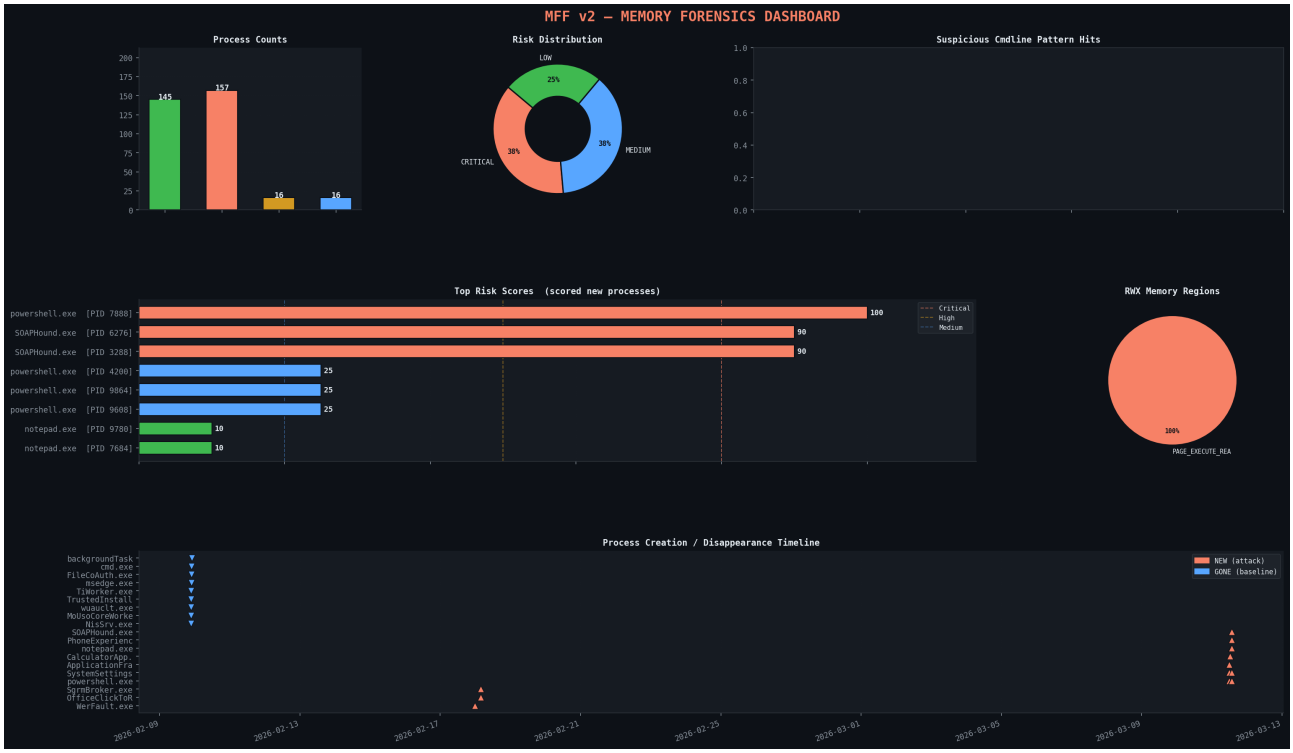
Defense Evasion

### IOC Summary:

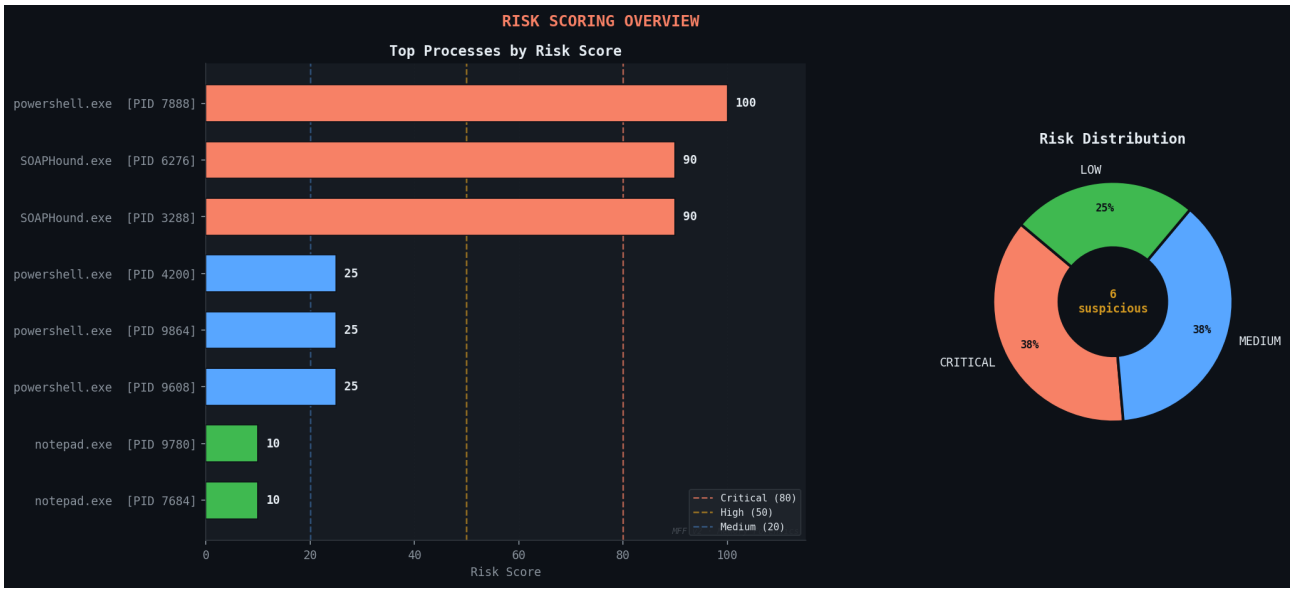
| Type | Indicators (top 8) |
|---|---|
| IPv4 | 0.0.0.0 · 13.107.213.40 · 13.89.179.9 · 172.183.7.192 · 172.183.7.194 · 184.31.68.248 |

# 2. VISUALIZATIONS

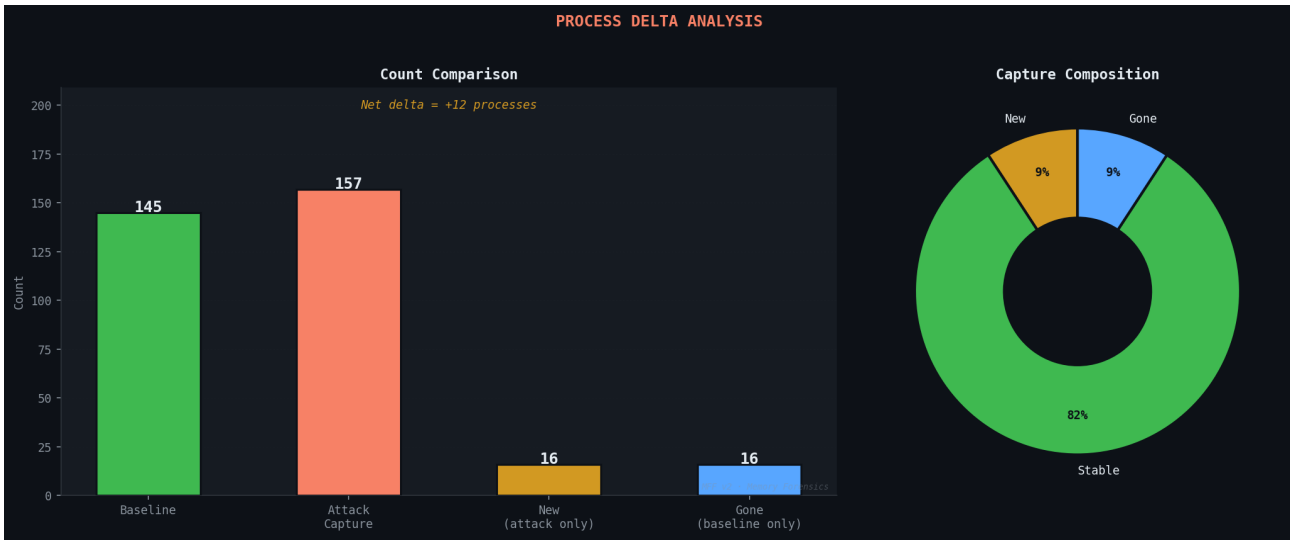
## 2.1 Full Forensics Dashboard



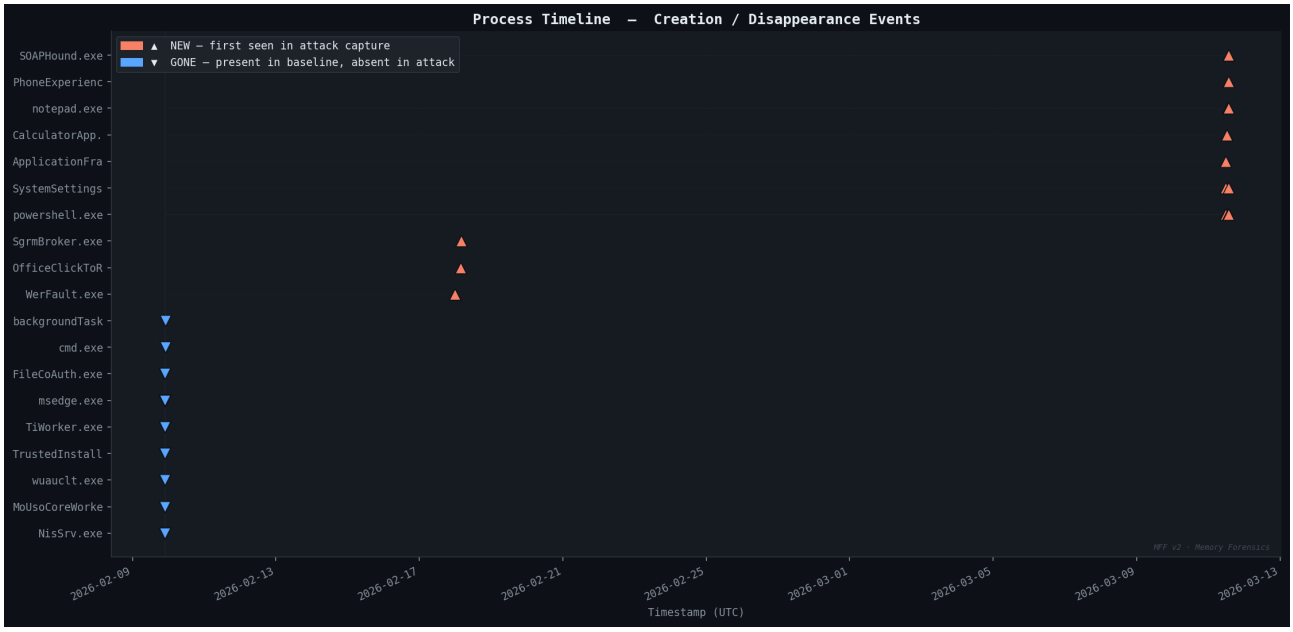
## 2.2 Risk Scoring Overview



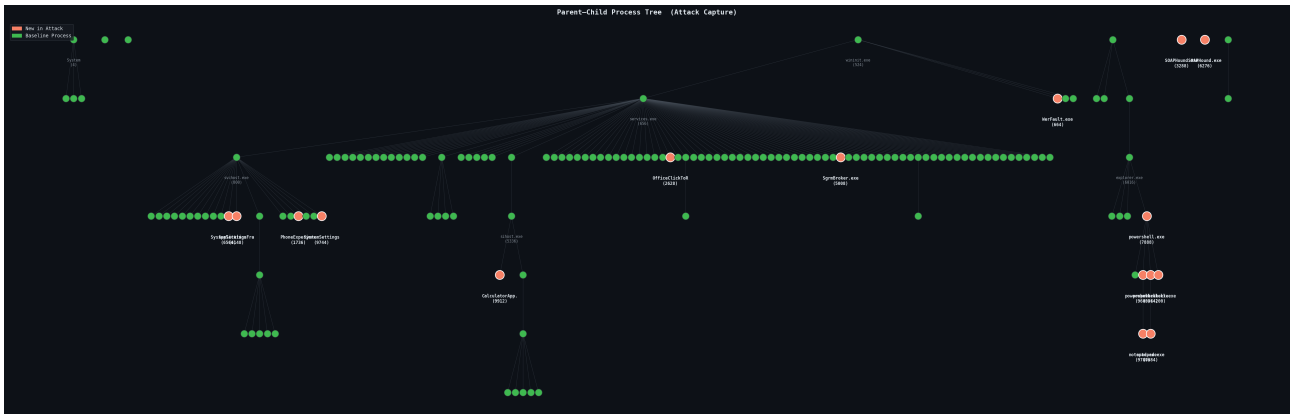
## 2.3 Process Delta Analysis



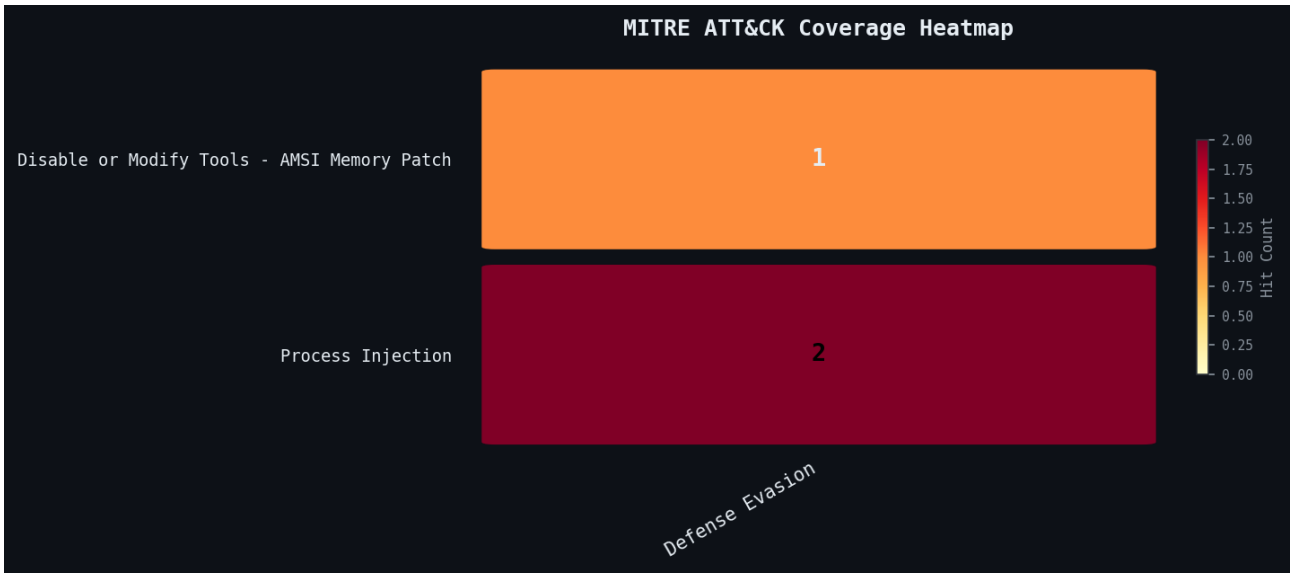

## 2.4 Process Timeline



## 2.5 Parent-Child Process Tree


Parent-Child Process Tree (Attack Capture)
New in Attack
Baseline Process
System
(4)
wininit.exe
(544)
services.exe
(684)
svchost.exe
(800)
SgrmBroker.exe
(5408)
MsMpEng.exe
(4016)
powershell.exe
(7988)
CalculatorApp.
(8922)
SDXHelperBackgroundHost.exe
(5208)
(8176)
WerFault.exe
(964)
OfficeClickToR
(2628)
(5104)
(1216)
(9704)
(5016)
(7880)
(7880)
(7880

## 2.6 MITRE ATT&CK; Heatmap




## 2.8 Malfind Memory Protections

## 3. RISK SCORE TABLE

| Process | PID | Score | Level | Indicators | | |
|---|---|---|---|---|---|---|
| powershell.exe | 7888 | 100 | CRITICAL | T1562.001 | 2 src | Medium-High |
| SOAPHound.exe | 6276 | 90 | CRITICAL | — | 1 src | Medium |
| SOAPHound.exe | 3288 | 90 | CRITICAL | — | 1 src | Medium |
| powershell.exe | 4200 | 25 | MEDIUM | — | 1 src | Medium |
| powershell.exe | 9864 | 25 | MEDIUM | — | 1 src | Medium |
| powershell.exe | 9608 | 25 | MEDIUM | — | 1 src | Medium |
| notepad.exe | 9780 | 10 | LOW | — | 1 src | Medium |
| notepad.exe | 7684 | 10 | LOW | — | 1 src | Medium |

## 4. NEW PROCESSES (Attack Only)

| ImageFileName | PID | PPID | CreateTime |
|---|---|---|---|
| WerFault.exe | 664 | 524 | 2026-02-17 23:49:52.000000 UTC |
| OfficeClickToR | 2628 | 656 | 2026-02-18 03:45:58.000000 UTC |
| SgrmBroker.exe | 5008 | 656 | 2026-02-18 03:48:05.000000 UTC |
| powershell.exe | 7888 | 6016 | 2026-03-11 11:48:11.000000 UTC |
| SystemSettings | 6564 | 800 | 2026-03-11 11:50:00.000000 UTC |
| ApplicationFra | 6148 | 800 | 2026-03-11 11:50:18.000000 UTC |
| CalculatorApp. | 9912 | 5336 | 2026-03-11 12:18:39.000000 UTC |
| powershell.exe | 9608 | 7888 | 2026-03-11 13:18:21.000000 UTC |
| notepad.exe | 9780 | 9608 | 2026-03-11 13:18:21.000000 UTC |
| powershell.exe | 9864 | 7888 | 2026-03-11 13:24:48.000000 UTC |
| notepad.exe | 7684 | 9864 | 2026-03-11 13:24:49.000000 UTC |
| PhoneExperienc | 1736 | 800 | 2026-03-11 13:28:34.000000 UTC |
| powershell.exe | 4200 | 7888 | 2026-03-11 13:29:21.000000 UTC |
| SOAPHound.exe | 6276 | 9816 | 2026-03-11 13:32:02.000000 UTC |
| SOAPHound.exe | 3288 | 9508 | 2026-03-11 13:32:03.000000 UTC |
| SystemSettings | 9744 | 800 | 2026-03-11 13:32:54.000000 UTC |

## 5. MITRE ATT&CK; TECHNIQUE MATCHES

| Technique | TechniqueName | Tactic | MatchedKeyword | PID | Process |
|---|---|---|---|---|---|
| T1055 | Process Injection | Defense Evasion | PAGE_EXECUTE_READWRITE | 7888 | powershell.exe |
| T1055 | Process Injection | Defense Evasion | EXECUTE_READWRITE | 7888 | powershell.exe |
| T1562.001 | Disable or Modify Tools - AMSI Memory Patch | Defense Evasion | amsi.dll | 7888 | powershell.exe |

## 6. SUSPICIOUS CMDLINE FINDINGS

No suspicious command lines found.

## 7. MALFIND — MEMORY INJECTION ARTEFACTS

| Process | PID | Protection |
|---|---|---|
| powershell.exe | 7888 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7888 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7888 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7888 | PAGE_EXECUTE_READWRITE |
| powershell.exe | 7888 | PAGE_EXECUTE_READWRITE |

## 7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

1 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

| PID | Process | DLL | HijackType | Technique | RiskScore | Tactic |
|---|---|---|---|---|---|---|
| 7888 | powershell.exe | amsi.dll | AMSI_BYPASS_MEMORY_PATCH | T1562.001 | 80 | Defense Evasion |

### [T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=7888)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=7888) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

## 8. NETWORK ARTEFACTS

### 8a. Flagged Connections (suspicious ports)

| TreeDepth | Offset | Proto | LocalAddr | LocalPort | ForeignAddr | ForeignPort | State | PID | Owner | Created | ImageFileName | SuspiciousPort | PortMeaning |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0xa4827a562e10 | TCPv4 | 0.0.0.0 | 135 | 0.0.0.0 | 0 | LISTENING | 920 | svchost.exe | 2026-02-17 23:49:54.000000 UTC | svchost.exe | 135 | DCOM/RPC |
| 0 | 0xa4827a563230 | TCPv4 | 0.0.0.0 | 135 | 0.0.0.0 | 0 | LISTENING | 920 | svchost.exe | 2026-02-17 23:49:54.000000 UTC | svchost.exe | 135 | DCOM/RPC |
| 0 | 0xa4827a563230 | TCPv6 | :: | 135 | :: | 0 | LISTENING | 920 | svchost.exe | 2026-02-17 23:49:54.000000 UTC | svchost.exe | 135 | DCOM/RPC |
| 0 | 0xa4827f8f3890 | TCPv4 | 10.0.2.5 | 139 | 0.0.0.0 | 0 | LISTENING | 4 | System | 2026-03-11 13:27:02.000000 UTC | System | 139 | NetBIOS |

## 9. IOC DETAIL TABLE

| Type | Value | Source | Count |
|---|---|---|---|
| FilePath | C:\\Windows\\system32\\svchost.exe | cmdline | 49 |
| FilePath | C:\\Windows\\System32\\svchost.exe | cmdline | 28 |
| FilePath | C:\\Program | cmdline | 12 |
| FilePath | C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView | cmdline | 7 |
| FilePath | Microsoft\\EdgeWebView\\Application\\144.0.3719.115\\msedgewebview2.exe | cmdline | 7 |
| FilePath | C:\\Windows\\System32\\RuntimeBroker.exe | cmdline | 6 |
| FilePath | C:\\ProgramData\\Microsoft\\Windows | cmdline | 2 |
| FilePath | C:\\Windows\\system32\\DllHost.exe | cmdline | 2 |
| FilePath | Microsoft | cmdline | 2 |
| FilePath | Windows | cmdline | 2 |
| FilePath | system32\\csrss.exe | cmdline | 2 |
| FilePath | ??\\C:\\Windows\\system32\\conhost.exe | cmdline | 1 |
| FilePath | C:\\Users\\lucif\\AppData\\Local\\Microsoft\\EdgeUpdate\\MicrosoftEdgeUpdate.exe | cmdline | 1 |
| FilePath | C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad | cmdline | 1 |
| FilePath | C:\\Windows\\Explorer.EXE | cmdline | 1 |
| FilePath | C:\\Windows\\ImmersiveControlPanel\\SystemSettings.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\SecurityHealthSystray.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\SystemSettingsBroker.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\VBoxService.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\VBoxTray.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe | cmdline | 1 |
| FilePath | C:\\Windows\\System32\\spoolsv.exe | cmdline | 1 |
| FilePath | C:\\Windows\\SystemApps\\Microsoft.Windows.Search_cw5n1h2txyewy\\SearchApp.exe | cmdline | 1 |
| FilePath | C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\explorer.exe | cmdline | 1 |

| Type | Value | Source | Count |
|---|---|---|---|
| FilePath | C:\\Windows\\system32\\ApplicationFrameHost.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\SearchIndexer.exe | cmdline | 1 |
| FilePath | C:\\Windows\\system32\\SecurityHealthService.exe | cmdline | 1 |

## 10. METHODOLOGY

**Acquisition:** VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

**Volatility 3:** Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

**Process diff:** Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

**Risk scoring:** Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

**Malfind filter:** Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

**DLL analysis:** windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

**ATT&CK; mapping:** 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

**IOC extraction:** Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

**Framework:** MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

**Risk scoring:** Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

**Standards:** ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.